Data Engineering Roadmap

A practical roadmap for becoming a professional Data Engineer

Recommended study schedule: 5 days/week × 3 hours/day = 15 hours/week ≈ 60 hours/month.

1. Programming Fundamentals

• Variables & data types

• Conditions

• Loops

• Functions/methods

• Arrays/lists

• OOP basics

• Exception handling

• File handling

• Git & GitHub

• **Language:** Python. Your Java knowledge is useful and does not need to be abandoned.

2. SQL — Very Important

• SELECT, WHERE, ORDER BY

• GROUP BY and HAVING

• JOINs

• Subqueries and CTEs

• Window functions

• CASE

• Views

• Indexes

• Constraints

• Transactions

• Query optimization

• Database normalization

• **Recommended databases:** PostgreSQL and MySQL.

3. Databases

• Relational databases

• Primary/foreign keys

• Relationships
• Normalization
• Transactions and ACID
• Indexes
• Query execution
• Partitioning
• Replication
• OLTP vs OLAP
• **Goal:** learn PostgreSQL deeply.

4. Linux + Command Line

• Linux filesystem
• cd, ls, cp, mv, rm
• Permissions
• Processes
• Environment variables
• SSH
• Bash scripting
• Cron jobs
• **Note:** Ubuntu/WSL is excellent practice for this.

5. Python for Data Engineering

• Python fundamentals
• Functions and OOP
• Modules/packages
• Virtual environments
• pip
• File handling
• JSON and CSV
• APIs
• Logging
• Error handling
• **Useful libraries:** pandas, requests, SQLAlchemy, psycopg, PySpark.

6. ETL / ELT — Core Skill

• ETL and ELT

• Data pipelines

• Batch processing

• Incremental loading

• Data validation

• Data quality

• Scheduling

• Retries and failure handling

7. Data Warehousing

• Data warehouse

• Data lake

• Data lakehouse

• OLTP vs OLAP

• Fact tables

• Dimension tables

• Star schema

• Snowflake schema

• Slowly Changing Dimensions

• Partitioning

• **Pick one warehouse first:** BigQuery, Snowflake, or Redshift.

8. Apache Spark

• Spark architecture

• RDD concept

• DataFrames

• Spark SQL

• Transformations

• Actions

• Partitioning

• Shuffling

• Caching

• Performance optimization

• **Recommended:** PySpark.

9. Workflow Orchestration — Airflow

• DAGs

• Tasks

• Operators

• Scheduling

• Dependencies

• Retries

• Monitoring

• XCom

• Connections

10. Cloud — Pick One First

• **Recommended:** AWS

• S3

• IAM

• EC2

• RDS

• Glue

• Lambda

• CloudWatch

• Redshift

• Later you can learn Azure or GCP.

11. Big Data & Streaming

• Distributed systems

• Streaming

• Message queues

• Event-driven architecture

• Batch vs streaming

• Apache Kafka

• Producers

• Topics

• Consumers

12. Data Engineering DevOps

- Git and GitHub
- Docker
- CI/CD
- Testing
- Environment variables
- Monitoring
- Logging
- Later: Kubernetes and Terraform

Complete Learning Order

Programming Fundamentals → Python → SQL → PostgreSQL / Databases → Linux + Bash → Git + GitHub → ETL / ELT → Data Warehousing → Airflow → PySpark → AWS → Kafka → Docker → Advanced Data Engineering

Portfolio Projects

Project 1 — CSV → Database

```
CSV files → Python → Clean data → PostgreSQL
```

Project 2 — API Pipeline

```
REST API → Python → Transform → PostgreSQL
```

Project 3 — Automated Pipeline

```
API → Python → Airflow → PostgreSQL
```

Project 4 — Big Data

```
Large dataset → PySpark → Transform → Data Warehouse
```

Project 5 — Real-World Cloud Pipeline

```
API → Python → AWS S3 → Spark → Warehouse → Dashboard
```

Estimated Timeline at 5 Days × 3 Hours

Stage	Approx. time
Python fundamentals	4–6 weeks
SQL + PostgreSQL	6–8 weeks
Linux + Git	2–3 weeks
ETL/ELT + pipelines	4–6 weeks
Data Warehousing	4–5 weeks
Airflow	3–4 weeks
PySpark	5–7 weeks
AWS	5–7 weeks
Kafka	3–5 weeks
Docker + advanced topics	3–4 weeks
Projects + revision	6–8 weeks

Expected total: about 10–14 months for the full roadmap.

Junior job-ready target: about 8–10 months if you consistently build projects and practice.

Strong foundation: about 12–14 months.

Priority for You

■ SQL

■ Python

■ Databases

4■■ Linux/Git

5■■ ETL

6■■ Airflow

7■■ Spark

8■■ Cloud

9■■ Kafka/Docker

Important: Java fundamentals are not wasted. They help you understand programming, OOP, loops, methods, and problem solving. For Data Engineering, however, Python and SQL should become your main focus.